

LABSHEET-4

1.Create a class Student with instance fields name, age and a class variable count to track number of students. Use local variables to assign values and display details.

```
class Student {  
  
    // Instance fields  
  
    String name;  
  
    int age;  
  
  
    // Class variable  
  
    static int count = 0;  
  
  
    // Constructor  
    Student(String name, int age) {  
        // Local variables are used as parameters  
  
        this.name = name;  
  
        this.age = age;  
  
        count++;  
    }  
  
  
    // Method to display student details  
  
    void displayDetails() {  
  
        System.out.println("Name: " + name);  
  
        System.out.println("Age: " + age);  
  
        System.out.println();  
    }  
  
  
    public static void main(String[] args) {  
  
        // Creating student objects
```

```
Student s1 = new Student("Varnika", 18);

Student s2 = new Student("Riya", 19);


// Display details

s1.displayDetails();

s2.displayDetails();


// Display total number of students

System.out.println("Total Students: " + count);

}

}
```

OUTPUT:-

Name: Varnika

Age: 18

Name: Riya

Age: 19

Total Students: 2

2) Create a class Rectangle with instance fields length, breadth, a class variable shapeName = "Rectangle". Write methods with local variables to calculate and display the area.

```
class Rectangle {

    // Instance fields

    double length;

    double breadth;


    // Class variable

    static String shapeName = "Rectangle";


    // Method to calculate and display area
```

```
void displayArea() {  
    // Local variable  
    double area = length * breadth;  
  
    System.out.println("Shape: " + shapeName);  
    System.out.println("Length: " + length);  
    System.out.println("Breadth: " + breadth);  
    System.out.println("Area: " + area);  
}  
  
public static void main(String[] args) {  
    // Create object  
    Rectangle r = new Rectangle();  
  
    // Assign values  
    r.length = 10;  
    r.breadth = 5;  
  
    // Display area  
    r.displayArea();  
}  
}
```

OUTPUT:-

Shape: Rectangle

Length: 10.0

Breadth: 5.0

Area: 50.0

3) Create a class Circle with instance field radius, a class variable PI = 3.14. Use local variables to calculate area and circumference.

```
class Circle {  
  
    // Instance field  
  
    double radius;  
  
  
    // Class variable  
  
    static double PI = 3.14;  
  
  
    // Method to calculate area and circumference  
  
    void calculate() {  
  
        // Local variables  
  
        double area = PI * radius * radius;  
  
        double circumference = 2 * PI * radius;  
  
  
        System.out.println("Radius: " + radius);  
  
        System.out.println("Area: " + area);  
  
        System.out.println("Circumference: " + circumference);  
    }  
  
  
    public static void main(String[] args) {  
  
        // Create object  
  
        Circle c = new Circle();  
  
  
        // Assign radius  
  
        c.radius = 5;  
  
  
        // Calculate and display
```

```
        c.calculate();  
    }  
}
```

OUTPUT:-

Radius: 5.0

Area: 78.5

Circumference: 31.400000000000002

4) Create a class Employee with instance fields emp Id, salary, and a class variable company Name. Display employee details using local variables.

```
class Employee {  
    // Instance fields  
  
    int empId;  
  
    double salary;  
  
    // Class variable  
  
    static String companyName = "ABC Company";  
  
    // Method to display employee details  
  
    void displayDetails() {  
        // Local variables  
  
        int id = empId;  
  
        double sal = salary;  
  
        System.out.println("Employee ID: " + id);  
  
        System.out.println("Salary: " + sal);  
  
        System.out.println("Company Name: " + companyName);  
    }  
}
```

```
public static void main(String[] args) {  
    // Create object  
    Employee e = new Employee();  
  
    // Assign values  
    e.empld = 101;  
    e.salary = 35000;  
  
    // Display details  
    e.displayDetails();  
}  
}
```

OUTPUT:-

Employee ID: 101

Salary: 35000.0

Company Name: ABC Company

5) Create a class Book with fields title, author, price, and a class variable publisher. Use local variables to set and display book details.

```
class Book {  
    String title;  
    String author;  
    double price;  
  
    static String publisher = "ABC Publications";  
  
    void displayDetails() {  
        String t = "Java Programming";  
        String a = "James Gosling";
```

```
double p = 450;
```

```
title = t;
```

```
author = a;
```

```
price = p;
```

```
System.out.println("Title: " + title);
```

```
System.out.println("Author: " + author);
```

```
System.out.println("Price: " + price);
```

```
System.out.println("Publisher: " + publisher);
```

```
}
```

```
public static void main(String[] args) {
```

```
    Book b = new Book();
```

```
    b.displayDetails();
```

```
}
```

```
}
```

OUTPUT:-

Title: Java Programming

Author: James Gosling

Price: 450.0

Publisher: ABC Publications

6) Create a class Car with fields brand, mileage, and a class variable wheels = 4. Display details
class Car {

```
    String brand;
```

```
    double mileage;
```

```
    static int wheels = 4;
```

```
void displayDetails() {  
    String b = "Toyota";  
    double m = 20.5;  
  
    brand = b;  
    mileage = m;  
  
    System.out.println("Brand: " + brand);  
    System.out.println("Mileage: " + mileage + " km/l");  
    System.out.println("Wheels: " + wheels);  
}
```

```
public static void main(String[] args) {  
    Car c = new Car();  
    c.displayDetails();  
}  
}
```

OUTPUT:-

Brand: Toyota

Mileage: 20.5 km/l

Wheels: 4

7) Create a class Bank Account with fields account Number, balance, and a class variable bank Name. Use local variables in deposit method.

```
class BankAccount {  
    String accountNumber;  
    double balance;
```

```
static String bankName = "State Bank";

void deposit() {
    double amount = 5000;

    balance = balance + amount;

    System.out.println("Bank Name: " + bankName);
    System.out.println("Account Number: " + accountNumber);
    System.out.println("Deposit Amount: " + amount);
    System.out.println("Balance: " + balance);
}

public static void main(String[] args) {
    BankAccount b = new BankAccount();

    b.accountNumber = "1234567890";
    b.balance = 10000;

    b.deposit();
}
}
```

OUTPUT:-

Bank Name: State Bank

Account Number: 1234567890

Deposit Amount: 5000.0

Balance: 15000.0

8) Create a class Pen with fields color, type, and a class variable manufacturer. Use local variables to display pen details.

```
class Pen {  
  
    String color;  
  
    String type;  
  
  
    static String manufacturer = "Cello";  
  
  
    void displayDetails() {  
  
        String c = "Blue";  
  
        String t = "Ball Pen";  
  
  
        color = c;  
  
        type = t;  
  
  
        System.out.println("Color: " + color);  
  
        System.out.println("Type: " + type);  
  
        System.out.println("Manufacturer: " + manufacturer);  
    }  
  
  
    public static void main(String[] args) {  
  
        Pen p = new Pen();  
  
        p.displayDetails();  
  
    }  
}
```

OUTPUT:-

Color: Blue

Type: Ball Pen

Manufacturer: Cello

9) Create a class Laptop with fields brand, RAM, and a class variable os = "Windows". Use local variables to display details.

```
class Laptop {  
    String brand;  
    int RAM;  
  
    static String os = "Windows";  
  
    void displayDetails() {  
        String b = "Dell";  
        int r = 8;  
  
        brand = b;  
        RAM = r;  
  
        System.out.println("Brand: " + brand);  
        System.out.println("RAM: " + RAM + " GB");  
        System.out.println("Operating System: " + os);  
    }  
  
    public static void main(String[] args) {  
        Laptop l = new Laptop();  
        l.displayDetails();  
    }  
}
```

OUTPUT:-

Brand: Dell

RAM: 8 GB

Operating System: Windows

10) Create a class Movie with fields name, genre, rating, and a class variable industry. Display movie details using local variables.

```
class Movie {  
  
    String name;  
  
    String genre;  
  
    double rating;  
  
  
    static String industry = "Bollywood";  
  
  
    void displayDetails() {  
  
        String n = "3 Idiots";  
  
        String g = "Comedy-Drama";  
  
        double r = 8.4;  
  
  
        name = n;  
  
        genre = g;  
  
        rating = r;  
  
  
        System.out.println("Movie Name: " + name);  
        System.out.println("Genre: " + genre);  
        System.out.println("Rating: " + rating);  
        System.out.println("Industry: " + industry);  
    }  
  
  
    public static void main(String[] args) {  
  
        Movie m = new Movie();
```

```
        m.displayDetails();  
    }  
}
```

11) Create a class Calculator with methods for add, subtract, multiply, divide using local variables. Add a class variable operations Count to count total operations done.

```
class Calculator {  
    static int operationsCount = 0;  
  
    void add() {  
        int a = 20;  
        int b = 10;  
        int result = a + b;  
        operationsCount++;  
        System.out.println("Addition: " + result);  
    }  
  
    void subtract() {  
        int a = 20;  
        int b = 10;  
        int result = a - b;  
        operationsCount++;  
        System.out.println("Subtraction: " + result);  
    }  
  
    void multiply() {  
        int a = 20;  
        int b = 10;  
        int result = a * b;
```

```
        operationsCount++;  
        System.out.println("Multiplication: " + result);  
    }
```

```
void divide() {  
    int a = 20;  
    int b = 10;  
    int result = a / b;  
    operationsCount++;  
    System.out.println("Division: " + result);  
}
```

```
public static void main(String[] args) {  
    Calculator c = new Calculator();  
  
    c.add();  
    c.subtract();  
    c.multiply();  
    c.divide();  
  
    System.out.println("Total Operations: " + operationsCount);  
}  
}
```

OUTPUT:-

Addition: 30

Subtraction: 10

Multiplication: 200

Division: 2

Total Operations: 4

12) Create a class Student Marks with fields marks1, marks2, marks3, and a class variable university Name. Use local variables to calculate average.

```
class StudentMarks {  
  
    int marks1;  
  
    int marks2;  
  
    int marks3;  
  
  
    static String universityName = "COER University";  
  
  
    void calculateAverage() {  
  
        int m1 = marks1;  
  
        int m2 = marks2;  
  
        int m3 = marks3;  
  
  
        int total = m1 + m2 + m3;  
  
        double average = total / 3.0;  
  
  
        System.out.println("Marks 1: " + m1);  
        System.out.println("Marks 2: " + m2);  
        System.out.println("Marks 3: " + m3);  
        System.out.println("University: " + universityName);  
        System.out.println("Average: " + average);  
    }  
  
  
    public static void main(String[] args) {  
  
        StudentMarks s = new StudentMarks();  

```

```
s.marks1 = 80;

s.marks2 = 75;

s.marks3 = 85;


s.calculateAverage();

}

}
```

OUTPUT:-

Marks 1: 80

Marks 2: 75

Marks 3: 85

University: COER University

Average: 80.0

13) Create a class Temperature with a field celsius and a class variable scale = "Celsius". Use local variables to convert into Fahrenheit and Kelvin.

```
class Temperature {

    double celsius;


    static String scale = "Celsius";


    void convertTemperature() {

        double c = celsius;

        double fahrenheit = (c * 9 / 5) + 32;

        double kelvin = c + 273.15;


        System.out.println("Temperature: " + c + " " + scale);

        System.out.println("Fahrenheit: " + fahrenheit);

    }

}
```

```
        System.out.println("Kelvin: " + kelvin);  
    }  
}
```

```
public static void main(String[] args) {  
    Temperature t = new Temperature();  
  
    t.celsius = 25;  
  
    t.convertTemperature();  
}  
}
```

OUTPUT:-

Temperature: 25.0 Celsius

Fahrenheit: 77.0

Kelvin: 298.15

14) Create a class Person with fields name, age and a class variable country. Use local variables to check voting eligibility.

```
class Person {  
    String name;  
    int age;  
  
    static String country = "India";  
  
    void checkEligibility() {  
        String n = name;  
        int a = age;  
  
        System.out.println("Name: " + n);  
    }  
}
```

```
System.out.println("Age: " + a);
```

```
System.out.println("Country: " + country);
```

```
if (a >= 18) {
```

```
    System.out.println("Eligible for Voting");
```

```
} else {
```

```
    System.out.println("Not Eligible for Voting");
```

```
}
```

```
}
```

```
public static void main(String[] args) {
```

```
    Person p = new Person();
```

```
    p.name = "Varnika";
```

```
    p.age = 18;
```

```
    p.checkEligibility();
```

```
}
```

```
}
```

OUTPUT:-

Name: Varnika

Age: 18

Country: India

Eligible for Voting

15) Create a class Simple Interest with fields principal, rate, time, and a class variable bank. Use local variables to calculate interest.

```
class SimpleInterest {
```

```
    double principal;
```

```
double rate;
```

```
double time;
```

```
static String bank = "State Bank";
```

```
void calculateInterest() {
```

```
    double p = principal;
```

```
    double r = rate;
```

```
    double t = time;
```

```
    double interest = (p * r * t) / 100;
```

```
    System.out.println("Bank: " + bank);
```

```
    System.out.println("Principal: " + p);
```

```
    System.out.println("Rate: " + r + "%");
```

```
    System.out.println("Time: " + t + " years");
```

```
    System.out.println("Simple Interest: " + interest);
```

```
}
```

```
public static void main(String[] args) {
```

```
    SimpleInterest s = new SimpleInterest();
```

```
    s.principal = 10000;
```

```
    s.rate = 5;
```

```
    s.time = 2;
```

```
    s.calculateInterest();
```

```
}  
}
```

OUTPUT:-

Bank: State Bank

Principal: 10000.0

Rate: 5.0%

Time: 2.0 years

Simple Interest: 1000.0

16) Create a class Triangle with sides a, b, c, and a class variable type "Scalene/Isosceles/Equilateral". Use local variables to check validity.

```
class Triangle {  
  
    double a;  
  
    double b;  
  
    double c;  
  
  
    static String type = "Scalene/Isosceles/Equilateral";  
  
    void checkValidity() {  
  
        double x = a;  
  
        double y = b;  
  
        double z = c;  
  
  
        if (x + y > z && x + z > y && y + z > x) {  
  
            System.out.println("Triangle is Valid");  
  
  
            if (x == y && y == z) {  
  
                System.out.println("Type: Equilateral");  
  
            } else if (x == y || y == z || x == z) {
```

```
        System.out.println("Type: Isosceles");
    } else {
        System.out.println("Type: Scalene");
    }
} else {
    System.out.println("Triangle is Not Valid");
}
}
```

```
public static void main(String[] args) {
    Triangle t = new Triangle();

    t.a = 5;
    t.b = 6;
    t.c = 7;

    t.checkValidity();
}
}
```

OUTPUT:-

Triangle is Valid

Type: Scalene

17) Create a class Student Grade with fields name, marks, and a class variable passing Marks. Use local variables to assign grade.

```
class StudentGrade {
    String name;
    int marks;
```

```
static int passingMarks = 40;
```

```
void assignGrade() {
```

```
    String n = "Varnika";
```

```
    int m = 85;
```

```
    name = n;
```

```
    marks = m;
```

```
    System.out.println("Name: " + name);
```

```
    System.out.println("Marks: " + marks);
```

```
    if (m >= passingMarks) {
```

```
        if (m >= 90) {
```

```
            System.out.println("Grade: A+");
```

```
        } else if (m >= 80) {
```

```
            System.out.println("Grade: A");
```

```
        } else if (m >= 70) {
```

```
            System.out.println("Grade: B");
```

```
        } else if (m >= 60) {
```

```
            System.out.println("Grade: C");
```

```
        } else {
```

```
            System.out.println("Grade: D");
```

```
        }
```

```
    } else {
```

```
        System.out.println("Grade: F");
```

```
    }
```

```
}
```

```
public static void main(String[] args) {  
    StudentGrade s = new StudentGrade();  
    s.assignGrade();  
}
```

```
}
```

OUTPUT:-

Name: Varnika

Marks: 85

Grade: A

18) Create a class Electric Bill with a field units and a class variable fixed Charge. Use local variables to calculate total bill.

```
class ElectricBill {  
    int units;  
  
    static double fixedCharge = 100;  
  
    void calculateBill() {  
        int u = units;  
        double charge;  
  
        if (u <= 100) {  
            charge = u * 5;  
        } else if (u <= 200) {  
            charge = (100 * 5) + ((u - 100) * 7);  
        } else {  
            charge = (100 * 5) + (100 * 7) + ((u - 200) * 10);  
        }  
    }  
}
```

```
}
```

```
double totalBill = charge + fixedCharge;
```

```
System.out.println("Units: " + u);
```

```
System.out.println("Fixed Charge: " + fixedCharge);
```

```
System.out.println("Total Bill: " + totalBill);
```

```
}
```

```
public static void main(String[] args) {
```

```
    ElectricBill e = new ElectricBill();
```

```
    e.units = 250;
```

```
    e.calculateBill();
```

```
}
```

```
}
```

OUTPUT:-

Units: 250

Fixed Charge: 100.0

Total Bill: 1800.0

19) Create a class Product with fields id, name, price, and a class variable discount Rate. Use local variables to calculate final price.

```
class Product {
```

```
    int id;
```

```
    String name;
```

```
    double price;
```

```
static double discountRate = 10;
```

```
void calculateFinalPrice() {
```

```
    int i = id;
```

```
    String n = name;
```

```
    double p = price;
```

```
    double discount = (p * discountRate) / 100;
```

```
    double finalPrice = p - discount;
```

```
    System.out.println("Product ID: " + i);
```

```
    System.out.println("Product Name: " + n);
```

```
    System.out.println("Price: " + p);
```

```
    System.out.println("Discount: " + discountRate + "%");
```

```
    System.out.println("Final Price: " + finalPrice);
```

```
}
```

```
public static void main(String[] args) {
```

```
    Product p = new Product();
```

```
    p.id = 101;
```

```
    p.name = "Laptop";
```

```
    p.price = 50000;
```

```
    p.calculateFinalPrice();
```

```
}
```

```
}
```

OUTPUT:-

Product ID: 101

Product Name: Laptop

Price: 50000.0

Discount: 10.0%

Final Price: 45000.0

using local variables.

20) Create a class Bank Account with fields balance, and a class variable bank Code. Use local variables to ensure withdrawal does not exceed balance.

```
class BankAccount {  
    double balance;  
  
    static String bankCode = "SBI001";  
  
    void withdraw() {  
        double amount = 5000;  
        double currentBalance = balance;  
  
        if (amount <= currentBalance) {  
            currentBalance = currentBalance - amount;  
            balance = currentBalance;  
  
            System.out.println("Bank Code: " + bankCode);  
            System.out.println("Withdrawal Amount: " + amount);  
            System.out.println("Remaining Balance: " + balance);  
        } else {  
            System.out.println("Bank Code: " + bankCode);  
            System.out.println("Insufficient Balance");  
        }  
    }  
}
```

```
}  
}
```

```
public static void main(String[] args) {  
    BankAccount b = new BankAccount();  
  
    b.balance = 10000;  
  
    b.withdraw();  
}  
}
```

OUTPUT:-

Bank Code: SBI001

Withdrawal Amount: 5000.0

Remaining Balance: 5000.03

21) Create a class Palindrome with a method using local variables to check if a number is palindrome. Add a class variable countChecks.

```
class Palindrome {  
    static int countChecks = 0;  
  
    void checkPalindrome(int num) {  
        int original = num;  
        int reverse = 0;  
        int digit;  
  
        while (num > 0) {  
            digit = num % 10;  
            reverse = reverse * 10 + digit;
```

```

        num = num / 10;
    }

    countChecks++;

    if (original == reverse)
        System.out.println(original + " is a Palindrome");
    else
        System.out.println(original + " is not a Palindrome");
}

public static void main(String[] args) {
    Palindrome p = new Palindrome();

    p.checkPalindrome(121);
    p.checkPalindrome(123);

    System.out.println("Total Checks: " + countChecks);
}
}

```

OUTPUT:-

121 is a Palindrome

123 is not a Palindrome

Total Checks: 2

22) Create a class Armstrong with a method using local variables to check Armstrong number. Add a class variable total Checks.

```

class Armstrong {
    static int totalChecks = 0;
}

```

```

void checkArmstrong(int num) {

    int original = num;

    int sum = 0;

    int digit;

    while (num > 0) {

        digit = num % 10;

        sum = sum + (digit * digit * digit);

        num = num / 10;

    }

    totalChecks++;

    if (original == sum)

        System.out.println(original + " is an Armstrong number");

    else

        System.out.println(original + " is not an Armstrong number");

}

public static void main(String[] args) {

    Armstrong a = new Armstrong();

    a.checkArmstrong(153);

    a.checkArmstrong(123);

    System.out.println("Total Checks: " + totalChecks);

```

```
}  
}
```

OUTPUT:-

153 is an Armstrong number

123 is not an Armstrong number

Total Checks: 2

23) Create a class Prime Checker with a method using local variables to check prime. Add a class variable total Prime Checks.

```
class PrimeChecker {  
    static int totalPrimeChecks = 0;  
  
    void checkPrime(int num) {  
        int count = 0;  
  
        for (int i = 1; i <= num; i++) {  
            if (num % i == 0)  
                count++;  
        }  
  
        totalPrimeChecks++;  
  
        if (count == 2)  
            System.out.println(num + " is a Prime number");  
        else  
            System.out.println(num + " is not a Prime number");  
    }  
  
    public static void main(String[] args) {
```

```
PrimeChecker p = new PrimeChecker();
```

```
p.checkPrime(7);
```

```
p.checkPrime(10);
```

```
System.out.println("Total Prime Checks: " + totalPrimeChecks);
```

```
}
```

```
}
```

OUTPUT:-

7 is a Prime number

10 is not a Prime number

Total Prime Checks: 2

24) Create a class Fibonacci with a method to display first N terms using local variables. Add a class variable series Count.

```
class Fibonacci {
```

```
    static int seriesCount = 0;
```

```
    void displaySeries(int n) {
```

```
        int a = 0;
```

```
        int b = 1;
```

```
        int c;
```

```
        for (int i = 1; i <= n; i++) {
```

```
            System.out.print(a + " ");
```

```
            c = a + b;
```

```
            a = b;
```

```
            b = c;
```

```
        }
```

```

        seriesCount++;

        System.out.println();
    }

    public static void main(String[] args) {

        Fibonacci f = new Fibonacci();

        f.displaySeries(7);

        f.displaySeries(5);

        System.out.println("Series Count: " + seriesCount);
    }
}

```

OUTPUT:-

0 1 1 2 3 5 8

0 1 1 2 3

Series Count: 2

25) Create a class Factorial with a method to compute factorial using local variables. Add a class variable total Calls.

```

class Factorial {

    static int totalCalls = 0;

    void calculateFactorial(int n) {

        int fact = 1;

        for (int i = 1; i <= n; i++) {

            fact = fact * i;

```

```

    }

    totalCalls++;

    System.out.println("Factorial of " + n + " = " + fact);
}

public static void main(String[] args) {
    Factorial f = new Factorial();

    f.calculateFactorial(5);
    f.calculateFactorial(4);

    System.out.println("Total Calls: " + totalCalls);
}
}

```

OUTPUT:-

Factorial of 5 = 120

Factorial of 4 = 24

Total Calls: 2

26) Create a class Matrix with fields for 2x2 matrix and a class variable matrix Type. Use local variables to add and subtract two matrices.

```

class Matrix {
    int[][] a = {{1, 2}, {3, 4}};
    int[][] b = {{5, 6}, {7, 8}};

    static String matrixType = "2x2";
}

```

```
void calculate() {  
  
    int[][] add = new int[2][2];  
  
    int[][] sub = new int[2][2];  
  
  
    for (int i = 0; i < 2; i++) {  
        for (int j = 0; j < 2; j++) {  
            add[i][j] = a[i][j] + b[i][j];  
            sub[i][j] = a[i][j] - b[i][j];  
        }  
    }  
  
  
    System.out.println("Matrix Type: " + matrixType);  
  
  
    System.out.println("Addition:");  
    for (int i = 0; i < 2; i++) {  
        for (int j = 0; j < 2; j++) {  
            System.out.print(add[i][j] + " ");  
        }  
        System.out.println();  
    }  
  
  
    System.out.println("Subtraction:");  
    for (int i = 0; i < 2; i++) {  
        for (int j = 0; j < 2; j++) {  
            System.out.print(sub[i][j] + " ");  
        }  
        System.out.println();  
    }  
}
```

```

    }
}

public static void main(String[] args) {
    Matrix m = new Matrix();
    m.calculate();
}
}

```

Output:

Matrix Type: 2x2

Addition:

6 8

10 12

27) Create a class StringReverse with a method using local variables to reverse a string. Add a class variable totalReversals.

```

class StringReverse {
    String text = "Hello";

    static int totalReversals = 0;

    void reverse() {
        String reversed = "";

        for (int i = text.length() - 1; i >= 0; i--) {
            reversed = reversed + text.charAt(i);
        }

        totalReversals++;
    }
}

```

```
        System.out.println("Original String: " + text);  
        System.out.println("Reversed String: " + reversed);  
        System.out.println("Total Reversals: " + totalReversals);  
    }  
}
```

```
public static void main(String[] args) {  
    StringReverse s = new StringReverse();  
    s.reverse();  
}  
}
```

Output:

Original String: Hello

Reversed String: olleH

Total Reversals: 1

28) Create a class Word Counter with a method using local variables to count words in a sentence. Add a class variable language.

```
class WordCounter {  
    String language = "English";  
  
    void countWords(String sentence) {  
        String[] words = sentence.trim().split("\\s+");  
        int count = words.length;  
  
        System.out.println("Sentence: " + sentence);  
        System.out.println("Language: " + language);  
        System.out.println("Number of words: " + count);  
    }  
}
```

```
public static void main(String[] args) {  
    WordCounter obj = new WordCounter();  
    obj.countWords("Java is easy to learn");  
}  
}
```

OUTPUT:-

Sentence: Java is easy to learn

Language: English

Number of words: 5

29) Create a class Employee with fields name, salary, and a class variable organization. Use local variables to compare salaries of two employees.

```
class Employee {  
    String name;  
    double salary;  
    static String organization = "ABC Company";  
  
    Employee(String n, double s) {  
        name = n;  
        salary = s;  
    }  
  
    void compareSalary(Employee e) {  
        double salary1 = salary;  
        double salary2 = e.salary;  
  
        System.out.println("Organization: " + organization);  
    }  
}
```

```
    if (salary1 > salary2)

        System.out.println(name + " has higher salary.");

    else if (salary2 > salary1)

        System.out.println(e.name + " has higher salary.");

    else

        System.out.println("Both employees have equal salary.");

}
```

```
public static void main(String[] args) {

    Employee e1 = new Employee("Rahul", 50000);

    Employee e2 = new Employee("Aman", 45000);


    e1.compareSalary(e2);

}

}
```

OUTPUT:-

Organization: ABC Company

Rahul has higher salary.

30) Create a class Library with fields books Available, and a class variable library Name. Use local variables to update issue and return operations.

```
class Library {

    int booksAvailable;

    static String libraryName = "City Library";


    Library(int books) {

        booksAvailable = books;

    }

}
```

```
void issueBook() {  
    int available = booksAvailable;  
  
    if (available > 0) {  
        available--;  
        booksAvailable = available;  
        System.out.println("Book issued successfully.");  
    } else {  
        System.out.println("No books available.");  
    }  
}
```

```
void returnBook() {  
    int available = booksAvailable;  
    available++;  
    booksAvailable = available;  
    System.out.println("Book returned successfully.");  
}
```

```
void display() {  
    System.out.println("Library: " + libraryName);  
    System.out.println("Books Available: " + booksAvailable);  
}
```

```
public static void main(String[] args) {  
    Library obj = new Library(5);
```

```
    obj.display();  
    obj.issueBook();  
    obj.returnBook();  
    obj.display();  
}  
}
```

OUTPUT:-

Library: City Library

Books Available: 5

Book issued successfully.

Book returned successfully.

Library: City Library

Books Available: 5